Examiner reports

Q1.

Although there were many correct force diagrams, there were also a significant number which included friction or other forces as well as the two required. A good number of candidates were able to find the acceleration when the slope was smooth, although there were a few candidates who did not show enough working, simply giving statements like "$a = \frac{1}{2}g$ = 4.9 m s^{-2}". In part (b), many candidates were able to find the acceleration successfully, but there were two common errors. The first was to assume that they could use = 5 v without any justification. The second was to simply divide the distance by the time. When finding the friction force, some candidates used $F = ma$ to obtain a friction force of 7.5 N. There were also a number of correct solutions. When finding the coefficient of friction, many candidates with incorrect values for the friction benefited from the follow through marks and gained full marks. There were however some cases where the normal reaction was incorrect either because no resolving had taken place or because the resolving had been incorrect.

In part (c), there were some good answers from candidates, but there were also quite a few which stated that the acceleration would be less, but did not give a good enough explanation for this.

Q2.

Parts (a), (b) and (c) of this question were often done very well. In part (a) candidates occasionally gave the answers "5" or "5**j**", but the vast majority of the answers were correct. In part (b), many correctly stated the velocity, but some candidates then performed incorrect simplifications. Part (c) was also done well by those candidates who had obtained the velocity vector.

Part (d) proved to be more challenging. Some candidates realised that they needed to find the position vector and hence the bearing. These candidates usually made good progress, although some stopped when they had the position vector and others gave the answer as 52.3°. Some candidates tried to work with velocities.

Q3.

There was considerable confusion here in all parts between the initial and final velocities and with the sign of the acceleration due to gravity. In part (b)(ii) it was not uncommon to see a new calculation instead of simply doubling the answer to part (b)(i) as intended.

Q4.

This question proved difficult, and a number of candidates showed a lack of appreciation of the differences between the horizontal and vertical components of the motion. Solutions in part (a) sometimes lacked convincing reasoning. Algebra let some down in part (c), and part (d) proved very discriminating on the understanding of the motion.

Q5.

Again there were frequent sign errors in part (a)(i). However, the rest of the question was popular and mostly done well with many fully complete solutions.

Q6.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

Q7.

Answers to part (a) were mostly good except for occasional sign errors. Part (b) revealed some misconceptions, with some candidates thinking the weights of the falling objects to be a significant factor in the motion. Many understood the effect of air resistance but were not always able to clearly express the link with the change to the motion.

Q8.

The sketch graph in part (a) proved a good source of marks although some graphs were drawn without the use of a ruler and intended straight lines were 'wobbly'. Part (a)(ii) was usually successful although some omitted some of the areas in trying to find the total distance. Parts (a)(iii) and (a)(iv) were popular and successful. Part (b)(i) was done well, but part (b)(ii) mostly lacked the required link between the curves of the graph and the more realistic changes in motion.

Q9.

A popular question with high marks awarded. It was pleasing that many scored full marks although some assumed the speed to be constant in part (b).

Q10.

A straightforward question, but not always yielding the marks it should. In part (a) many were careless with working, often confusing values of u and v, and altering or ignoring incorrect signs in subsequent working. Candidates should be aware that 'show that' requests require convincing correct working leading to the printed result. In part (b) sketches were usually good but sometimes marred by incomplete labelling. Part (c) was done well. Responses in part (d) sometimes referred to irrelevant factors rather than the model of the motion given in the question.

Q11.

This was found to be the most difficult question on the paper. While the most able were able to score full marks, others showed confusion between vectors and scalars, or simply showed misunderstanding of vector quantities. Many found the vector $\dot{AB}$ in part (a) but could not then proceed correctly, with ratios of vectors often leading to vector expressions for the time. The most successful methods included the application of the equations of motion with constant acceleration, and the use of diagrams with explained steps in motion. Part (b) often began with the assumption of a zero initial velocity, some only considered the **j** component of the motion, and many failed to add the position vector of the particle at B. Part (c) often began well with the vector $\overline{AC}$ found, but with no attempt to find its magnitude to give the distance requested.

Q12.

This question proved quite challenging, with the least successful candidates confusing the horizontal and vertical components of motion, and in some cases trying to incorporate expressions for the greatest height, horizontal range, and time of flight, thereby making the question harder. Part (a) was mostly done well, but some assumed vertical motion was involved. Part (b) was sometimes omitted, while a significant number did not realise that the calculation could be done in one stage and wasted time in carrying out a series of calculations often losing an accuracy mark due to rounding values within their working. In part (c)(i) common errors included finding the vertical component of velocity only, using '24' as the initial velocity with no component, and mixing horizontal/vertical motions or velocity/displacement in calculations. Those successful in part (c)(i) usually completed part (c)(ii).

Q13.

A popular question with high marks awarded. It was pleasing that many found the average speed as requested although some found the final speed instead.

Q14.

A straightforward question, but not always yielding the marks it should. In part (a) many candidates were careless with working, often confusing values of u and v, and altering or ignoring incorrect signs in subsequent working. Candidates should be aware that 'show that' requests require convincing correct working leading to the printed result. In part (b) sketches were usually good but sometimes marred by incomplete labelling. Part (c) was done well. Responses in part (d) sometimes referred to irrelevant factors rather than the model of the motion given in the question.

Q15.

Again diagrams were mostly good, and parts (a)(ii) and (c) usually gained full marks. Part (b) was successfully completed by most candidates with the main error being the inclusion of the weight of P instead of the frictional force in the equation of motion of P. Pleasingly very few candidates used a 'whole string' approach instead of forming equations of motion for the two particles as requested. In part (d) the intended response focussed on the remaining force acting on each particle and the subsequent change in speed. Many candidates were successful here but others focussed on forces no longer present.

Q16.

Part (a) The vast majority of candidates were familiar with the equations of motion for a projectile. They were able to eliminate t and arrive at the required equation of the trajectory.

Part (b) The algebraic manipulation and 'tidying up' of the quadratic equation in x and the subsequent task of solving it were prone to errors for some candidates. Some candidates who solved the equation correctly did not identify the required answer from their two solutions.

Part (c) Most candidates stated the two assumptions; no air resistance and the ball is a particle. A small minority of the candidates mistakenly stated that the ball was a particle without that mass.

Q17.

Most candidates did well on this question; almost all were able to obtain the acceleration without difficulty. Finding the distance was also done well, but a few candidates made arithmetic errors. There was also a small number who assumed a constant velocity. Part (b) caused a few more problems. Some candidates had trouble with the units and did not convert tonnes to kg. A few also made sign errors, while some others simply calculated the product of the mass and acceleration.

Q18.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

Q19.

Generally this question was done very well. An error that was observed several times in part (a) was to find the area above the graph, rather than the area below the graph. Part (b) was done very well. Part (c) was also done well, but some candidates rounded their final answer to two significant figures.

Q20.

Generally the candidates did well on his question. In part (a) there were a relatively small number of candidates who ignored the instruction to form an equation of motion for each particle. Some candidates used a method that produced $-2.8\ \mathrm{ms}^{-2}$. This approach was acceptable for part (a), but these candidates then used 2.8 in their equation to find the tension and obtained an incorrect value. In part (c), there were many very good responses. Some candidates failed to double their answer and lost the final mark. A few candidates worked with the acceleration due to gravity, rather than the acceleration that they had calculated in part (a).

Q21.

The candidates seemed to find the fact that the arrow was fired horizontally difficult to handle. Many appeared to try to give the arrow a vertical component of velocity. In part (a) many candidates seemed to have been helped by the printed answer. In some cases they included an unnecessary negative sign, which led to a negative time. There were some good answers to part (b), but some candidates included an acceleration g. There were very varied answers to part (c). Some candidates gave very good responses, but there were very many confused ones too, which revealed a lack of understanding of the principles of this topic.

Q22.

There were many good responses to this question. Generally part (a) was done very well. Part (b) was most demanding, but many candidates still did well on this part. One error was to work with a position vector instead of a velocity vector. In part (c), a few candidates gave the velocity as their final answer instead of the speed.

Q23.

Part (a) was often done well, but a number of candidates effectively repeated the same assumption twice. The candidates were often helped by the printed answer in part (b). Many of the candidates made life difficult for themselves by finding the initial speed and angle of projection and then working with these values. One consequence of this was that the final answers were inaccurate. Part (c) proved more challenging for candidates. The most common incorrect approach was to assume that the ball hit the wall when it was at its maximum height. In part (d) there were some good answers and many of the candidates were on the right track. However quite of few of the candidates produced answers which lacked clarity of explanation.

Q24.

This was done well by most, although there was occasional confusion seen in using the appropriate distance for parts (b) and (c).

Q25.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q26.

There were some excellent solutions to this question, which was intended to discriminate. The majority knew and attempted to apply the required principles but some were let down by poor algebraic skills in parts (a) and (b). Part (c) showed limited understanding of the equilibrium of the peg. Part (d) was mostly done well, but some failed to appreciate the need to double the distance found when considering the motion of one particle. It should be noted that in a totally algebraic question, candidates should be aware that a numerical value for g is inappropriate. However, the mark scheme treated candidates who used a numerical value sympathetically where possible.

Q27.

This question proved both popular and successful for candidates. Part (a) was very well done and part (b) only slightly less so with occasional accuracy errors or misunderstanding regarding the average speed.

Q28.

This question produced a variety of responses. Parts (a)(i) and (ii) were mostly answered well, although some showed all the forces acting rather than those on particle A as requested. Part (a)(iii) produced irrelevant work, as many failed to appreciate the equilibrium situation, and produced equations of motion as if a dynamic situation existed. Part (a)(iv) proved testing, but many showed a good understanding of the forces involved. Part (b) was generally done well and accurately.

Q29.

Candidates produced good work in this question, and were able to score marks on the parts where they had the necessary knowledge and understanding. Part (a)(i) was done well, and part (a)(ii), although an unfamiliar request, produced many good solutions. Part (a)(iii) was done well in a variety of ways. Parts (b) and (c) proved successful for most candidates.

Q30.

This question proved to be very demanding. Many candidates failed to understand the motion of the particle, not appreciating that its deceleration would cause the direction of motion to eventually reverse. There were consequently many erroneous attempts in part (a). Candidates who had found two velocities in part (a) were then able to gain marks in part (b), provided their values were consistent with the nature of the motion. There were, however, many elegant and concise solutions to both parts of the question.

Q31.

It was refreshing to see a wide variety of approaches to this question, which differed in many ways from previous questions on this topic. Many candidates were able to find the coordinates of the ball in part (a) and at this stage an error in the direction of the y-value was condoned. Part (b) provided interesting responses, using either the answers to part (a) with subsequent elimination of the time, or, in some cases, the full equation of the trajectory. In part (c) many candidates used the result of part (b), while others successfully found the time involved as an intermediate step. Part (d) proved to be more discriminating and there was confusion, for some, as to the quantities involved in the motion in the two directions. Many candidates reached the correct result following sound work, but some lost a mark through failing to adhere to the degree of accuracy required by the rubric for the paper. Overall it is encouraging to see candidates who are unable to complete one part of a question continue to other parts where they can be more successful.

Q32.

This question was done quite well, although some did not find the magnitude of the appropriate quantity in part (a), while others conveniently changed signs in the midst of working to obtain a positive answer.

The simplicity of the request in part (b) was not always realised, and in part (c), some could not find the normal reaction force correctly.

Q33.

The majority of candidates were successful in part (a) but too many could not attempt part (b) with an appropriate method. Part (c) was sometimes omitted, and some criticisms did not relate to the model given.

Q34.

A surprising number found the static part of this question very demanding, with relatively few correct responses to parts (a)(i), (iii) and (iv). Part (b) proved more successful.

Q35.

Parts (a) and (b) were done well by the better prepared candidates, but few spotted and used the pattern of decreasing distances in part (c).